

## 3.6 Libraries in Python (Built-in & Third-Party)

لائبریری پہلے سے لکھا ہوا کوڈ ہے جسے import سے پروگرام میں لایا جاتا ہے۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 12 of 13

### 3.6 Libraries in Python (Built-in & Third-Party)

#### LEARN IT

A **library** (also called a **module**) is a collection of pre-written code that offers ready-made functions. Instead of writing a square-root algorithm from scratch, you import `math` and call `math.sqrt()`. Libraries save time, reduce bugs and let you stand on other programmers' work.

There are five common ways to import a library. Method 3 (import only what you need) is the best habit: it is clearer and uses less memory than `from math import *`.

| Method               | Syntax                                 | How you call the function          |
|----------------------|----------------------------------------|------------------------------------|
| 1. Entire module     | <code>import math</code>               | <code>math.sqrt(25)</code>         |
| 2. Alias (nickname)  | <code>import math as m</code>          | <code>m.sqrt(16)</code>            |
| 3. Specific function | <code>from math import sqrt</code>     | <code>sqrt(36)</code>              |
| 4. Several functions | <code>from math import sqrt, pi</code> | <code>sqrt(9), pi</code>           |
| 5. Everything (*)    | <code>from math import *</code>        | <code>sqrt(64)</code> — avoid this |

### WORKED EXAMPLE

Five imports, one result each:

```
import math
print(math.sqrt(25))
import math as m
print(m.sqrt(16))
from math import sqrt
print(sqrt(36))
from math import sqrt, pi
print(pi)
from math import *
print(sqrt(64))
```

#### OUTPUT

```
5.0
4.0
6.0
3.141592653589793
8.0
```

### COMMON MISTAKE

`from math import *` dumps every name into your program. If you later write a function called `sqrt`, you silently overwrite the library one. Prefer methods 1–4.

### اردو میں سمجھیں

لائبریری پہلے سے لکھا ہوا کوڈ ہے جسے `import` سے پروگرام میں لایا جاتا ہے۔ سب سے اچھا طریقہ یہ ہے کہ صرف وہ فنکشن درآمد کریں جس کی ضرورت ہو، ستارہ `*` سے سب کچھ نہ کھینچیں۔

### CHECK YOURSELF

1. Write the line that lets you call `ceil` without writing `math.` in front.
2. Why is an alias such as `import pandas as pd` useful?

Answers: 1. `from math import ceil`. 2. The library name is long; a short nickname keeps later lines short and is a community convention.

### LEARN IT

**Built-in libraries** come pre-installed with Python. You only import them; you do not run `pip`. Three that every Class XI student must know are `math`, `random` and `datetime`.

## math

```
import math
print(math.pi)
print(math.ceil(4.3))
print(math.floor(4.9))
print(math.pow(2, 4))
print(math.sqrt(81))
```

### OUTPUT

```
3.141592653589793
5
4
16.0
9.0
```

## random and datetime

```
import random
print(random.randint(1, 1))
print(0.0 ≤ random.random() < 1.0)

import datetime
now = datetime.datetime(2026, 9, 27,
11, 59, 40)
print(now.strftime("%d-%m-%Y"))
```

### OUTPUT

```
1
True
27-09-2026
```

`randint(a, b)` includes both ends. `random()` is a float in `[0.0, 1.0)`. `strftime` formats a date; `%d-%m-%Y` is day-month-year.

### REMEMBER IT

`ceil` always rounds **up** (ceiling); `floor` always rounds **down**. `round()` (built-in, not in `math`) goes to the nearest integer and is a different function.

### اردو میں سمجھیں

`math` حساب کے لیے، `random` بے ترتیب عدد کے لیے اور `datetime` تاریخ وقت کے لیے بنی ہوئی لائبریریاں ہیں۔ انہیں انسٹال نہیں کرنا پڑتا، صرف `import` کافی ہے۔

### CHECK YOURSELF

1. What does `math.ceil(4.0)` return?
2. Write a line that prints a random integer from 1 to 6 (a dice).

Answers: 1. 4 (already an integer). 2. `print(random.randint(1, 6))` after `import random`.

### LEARN IT

**Third-party libraries** are written by the community. They are **not** bundled with Python, so you install them from a terminal with `pip install library_name` and then import them as usual.

| Library | Primary use                                |
|---------|--------------------------------------------|
| NumPy   | Heavy numerical computing and large arrays |

|                      |                                              |
|----------------------|----------------------------------------------|
| Pandas               | Data analysis; table-like data (CSV / Excel) |
| Matplotlib           | Data visualisation — graphs and charts       |
| TensorFlow / PyTorch | Artificial intelligence and machine learning |

### WORKED EXAMPLE

Installing and importing Pandas (you will use this in Class XII):

```
pip install pandas
import pandas as pd
```

### COMMON MISTAKE

`pip` is a **terminal** command, not a Python statement. Do not type `pip install pandas` inside a `.py` file. In VS Code, open the Terminal panel.

### اردو میں سمجھیں

تھرڈ پارٹی لائبریریاں پائتھن کے ساتھ نہیں آتیں، انہیں `pip install` سے انسٹال کیا جاتا ہے۔ نمپائی، پینڈاز، میٹپلاٹ لیب اور ٹینسر فلو اسی قسم کی ہیں۔

### CHECK YOURSELF

1. Which library would you choose to draw a bar chart of class marks?
2. Is `math` a third-party library?

Answers: 1. Matplotlib. 2. No — `math` is built-in.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** A library is:

- a. Pre-written functions you import
- b. A logic gate
- c. A cable
- d. A sprint

**MCQ 2.** math is:

- a. Built-in / standard library
- b. Third-party only
- c. A DBMS
- d. An OSI layer

**MCQ 3.** pandas is:

- a. Third-party (pip install)
- b. Built into every tiny Python
- c. A sort algorithm
- d. A gate

**MCQ 4.** import math then square root is:

- a. `math.sqrt(x)`
- b. `sqrt x`
- c. `x**` without import always for sqrt
- d. `print.sqrt`

**MCQ 5.** `random.randint(1,6)` is for:

- a. A dice integer 1–6
- b. Sorting
- c. Files
- d. TCP

**MCQ 6.** pip installs:

- a. Third-party packages
- b. The CPU
- c. OSI layer 1
- d. Urdu fonts only

## Extra short questions

1. Differentiate built-in and third-party libraries.
2. How do you call a library function?
3. Give two built-in and two third-party examples from this course.
4. Why libraries?

## Extra long question

1. Write a tiny program using math and random, then explain how you would add pandas (install + import). Mention a mistake: importing before installing.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

### Short

1. Built-in ship with Python (math, random, sqlite3). Third-party need pip (pandas, matplotlib).
2. `import lib; lib.func(...)` or `from lib import func`.
3. Built-in: math, random (or sqlite3 in XII). Third: pandas, matplotlib.
4. Do not reinvent sqrt, CSV readers, or plots.

### Long

1. `import math, random; print(math.pi, random.randint(1,6))`. Terminal: `pip install pandas`. Then import pandas as `pd`. `ImportError` if pip was skipped.

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Next lecture: 3.7 Debugging Code in Python